

BOILER TUBE FAILURE CASE STUDIES

Structured Root Cause Analysis of Six Recorded Tube Failures

DSPM Thermal Power Station, CSPGCL, Korba East - 2 x 250 MW

Units 1 and 2, January to July 2026

Prepared from the station's Tripping Analysis Reports and Executive Engineer (Boiler) failure analysis letters

Prepared: August 2026

1. Source records and how they are marked

This analysis is built from actual station documentation, not from illustrative examples. The source records supplied were:

- Tripping Analysis Report, Unit 1, tripped 25.01.2026 at 10:00 hrs - LTSH and economiser, pin hole, 32 joints.
- Tripping Analysis Report, Unit 2, tripped 19.06.2026 at 16:45 hrs - LTSH lower bank coil 131, burst open, 11 joints.
- Failure analysis letter no. 025-07/SE/BLR/BMD/U-I/4 - 2884 dated 12.07.2026 - Unit 1 leakage of 11.07.2026 at 21:17 hrs, economiser middle at 37 m / steam-cooled wall LHS door bend, pin hole, 6 joints.
- Tripping Analysis Report, Unit 1, tripped 14.07.2026 at 11:15 hrs, and the corresponding failure analysis letter no. 025-07/SE/BLR/BMD/U-I/5 dated 15.07.2026 - platen superheater inlet inner loop, burst open, 2 joints.

Throughout this document:

- [RECORD] marks information taken directly from those documents. It is reproduced as the station recorded it, including the recorded root cause.
- [ANALYSIS] marks engineering inference applied to those records. It is not part of the station's written finding and should be read as commentary offered for review, not as an established fact about the plant.

This separation matters because in one of the six cases the analysis reaches a different conclusion from the station's recorded root cause, and the reader needs to see clearly where the record ends and the interpretation begins.

2. How four trips become six cases

The six images supplied contain four distinct incidents (one report appears twice, and one incident is represented by both a Tripping Analysis Report and a failure analysis letter). Those four incidents contain six distinct failure locations, because two of the incidents each damaged two separate parts of the boiler. The cases are therefore organised by failure location rather than by trip, since each location has its own mechanism, its own evidence and its own remedy:

Case	Date	Unit	Zone and location of failure	Type	Joints	Recorded root cause
1	25.01.2026	Unit 1	LTSH lower bank coil 117 circuit 2 connection tubes, 41 m	Pin hole	32 (both zones)	Flue gas erosion
2	25.01.2026	Unit 1	Economiser upper coil 157, tube 9 from top (same trip)	Pin hole	8 of the 32	Flue gas erosion
3	19.06.2026	Unit 2	LTSH lower bank coil 131, tube 3 from bottom, 41 m	Burst open	11	Flue gas erosion
4	11.07.2026	Unit 1	Economiser middle 37 m - steam-cooled wall LHS door bend	Pin hole	6 (total)	Flue gas erosion
5	11.07.2026	Unit 1	Adjacent door bend and straight tube (extent of damage)	Secondary	part of 6	Flue gas erosion
6	14.07.2026	Unit 1	Platen superheater (inlet) inner loop	Burst open	2	Flue gas erosion

3. Recorded cost of these failures

Event	Unit	Outage (hrs)	Gen loss (MU)	Oil consumed	Load at trip
25.01.2026	Unit 1	31.70	7.925	23 KL LDO	85 MW
19.06.2026	Unit 2	25.00	6.250	16 KL LDO + 16 KL HSD	103 MW
11.07.2026	Unit 1	approx. 20	not recorded	not recorded	not recorded
14.07.2026	Unit 1	32.83	8.221	43 KL LDO	102 MW
Total (recorded)	-	89.53+	22.396+	82 KL LDO + 16 KL HSD	-

[ANALYSIS] Three of the four events were on Unit 1, and two of those fell within three days of each other. Every event occurred at 85-103 MW against a 250 MW rating, with three or four mills in service - the whole failure history sits in a deep part-load operating regime.

4. Analysis method

Each case answers the same eleven questions. Evidence is listed in the order it became available; several competing hypotheses are generated before any is preferred; each is tested against the evidence; the single most discriminating observation is identified; competing hypotheses are ruled out with stated reasoning; and the root cause is obtained by asking why the failure mechanism was permitted to reach the point of failure.

Where the record does not contain the evidence needed to discriminate between hypotheses, this is stated rather than glossed over. Case 6 is the clearest instance.

Case 1 - Unit 1, 25.01.2026: LTSH lower bank coil no. 117 connection tube pin-hole leak

1. Asset context

Asset model and make: DSPM TPS (Dr. Shyama Prasad Mukherjee Thermal Power Station), CSPGCL, Korba East - 2 x 250 MW coal-fired units. Boiler maker reference CG-489 (as cited in the station's boiler tube leakage failure analysis letters). Natural-circulation, drum-type, pulverised-coal fired boiler with mills designated A to E, radiant and platen superheater, LTSH lower and upper banks, economiser upper and middle banks, and steam-cooled wall enclosure to the second pass. Governing mode EHG.

Operating context: Korba East fires high-ash, high-quartz Indian coal from the Korba coalfield, which is the dominant driver of fly-ash erosion in the low-temperature convective zones (economiser, LTSH lower bank and their connection tubes). All four recorded trips occurred with the unit loaded in the range 85-103 MW against a 250 MW rating, i.e. deep part load, with only 3-4 mills in service. [RECORD] Parameters just before tripping: load 85 MW, drum pressure 72.59 kg/cm², main steam temperature 529 degC, mills in operation B, C, D and E, frequency 50.06 Hz, governor mode EHG, condenser vacuum 0.0731 kg/cm². The unit was therefore running at approximately 34% of rated load with four mills, well below the design operating point.

2. Incident summary - what happened

[RECORD] Unit 1 tripped on 25.01.2026 at 10:00 hrs and was stopped due to boiler tube leakage. At the ECR the window annunciation showed generator breaker auto-tripped; relays observed were GRP1 reverse power protection 37G1 with 186A and 186AX, GRP2 multiplier to 186C and 186CX, GRP3 Class A trip 286A with multiplier to 286A and 286AX, and GRP4 Class C trip 286C with multiplier to 286C and 286CX. Two zones of failure were recorded in the same event: the LTSH and the economiser. This case addresses the LTSH zone. The failure was located at LTSH lower bank coil no. 117, circuit 2 connection tubes at the 41 m elevation, 2 joints. The type of failure was recorded as a pin hole and the root cause of failure was recorded as failure due to flue gas erosion. The extent of damage covered adjacent connection tubes on coil no. 117 circuits 1 and 3 (4 joints), coil no. 116 circuits 1, 2 and 3 (6 joints), coil no. 115 circuits 1, 2 and 3 (6 joints), coil no. 118 circuits 1 and 2 (4 joints) and coil no. 119 circuit 1 (2 joints). Across both zones, 32 joints were welded and the failed tubes were replaced with new straight tube of 47.63 x 5.6 mm SA213 T11. The unit was lit up on 26.01.2026 at 14:00 hrs, synchronised at 17:42 hrs, with 31.7 hours outage, 7.925 MU generation loss and 23 KL LDO consumed.

2.1 Timeline and onset

- [RECORD] 25.01.2026, 10:00 hrs - Unit stopped due to boiler tube leakage; generator breaker auto-tripped; reverse power protection operated as the unit decelerated after the boiler was taken off.
- [ANALYSIS] The trip sequence recorded (boiler tube leakage leading to unit stoppage, then generator breaker trip on reverse power) indicates a controlled stoppage in which the boiler was tripped first and the generator followed on reverse power, rather than an instantaneous protection trip. This implies the leak was detected and acted upon by the operating staff.
- [ANALYSIS] Onset: a pin-hole leak of this type does not appear instantaneously. Wall loss by fly-ash erosion accumulates over thousands of operating hours; the pin hole is the moment the remaining wall can no longer contain the internal pressure. The detectable onset - rising make-up water, falling boiler water conductivity, audible steam noise in the second pass - would have preceded the trip by hours.
- [RECORD] 26.01.2026, 14:00 hrs - Boiler lit up after repair. 23 KL LDO consumed for the light-up.
- [RECORD] 26.01.2026, 17:42 hrs - Unit synchronised. Total outage 31.7 hours; generation loss 7.925 MU.
- [ANALYSIS] The 32 welded joints across the two zones indicate a repair scope far larger than the single leaking tube - the majority of the work was pre-emptive replacement of neighbouring eroded tubes, which is itself evidence of widespread, not localised, wall loss.

3. Evidence, in order of observation

- [RECORD] Unit stopped at 10:00 hrs due to boiler tube leakage - the primary observation logged at the MCR.
- [RECORD] Generator breaker auto-tripped with reverse power protection 37G1 operating - consistent with the boiler being removed from service first.
- [RECORD] Failure type recorded as a pin hole, not a burst - the failure morphology.
- [RECORD] Failure located at the connection tubes of LTSH lower bank coil 117 circuit 2 at 41 m, and a second failure in the economiser upper bank in the same event.
- [RECORD] Extent of damage covering five adjacent coils (115, 116, 117, 118, 119) - the damage was not confined to the failed tube.
- [RECORD] Root cause recorded by the station as failure due to flue gas erosion.
- [RECORD] Remedial action included protection shields provided at LTSH lower bank connection tubes, economiser lower bank and the door of the economiser middle zone - the station itself identified erosion protection as the remedy.
- [RECORD] Refractory applied at header 9 - indicating a gas-path bypass or refractory loss at that header.
- [ANALYSIS] Operation at 85 MW on a 250 MW unit with four mills - a deep part-load condition that alters gas-path velocity distribution.

4. Components affected by the failure

- [RECORD] LTSH lower bank coils 115, 116, 117, 118 and 119 and their circuit connection tubes at the 41 m elevation.
- [RECORD] Economiser upper bank (the second failure zone in the same event - treated separately in Case 2).
- [RECORD] Refractory at header 9.
- [ANALYSIS] Adjacent tubes exposed to steam-jet cutting from the escaping leak - the reason a pin hole in one tube produces damage across five coils.
- [ANALYSIS] Tube supports, spacers and alignment in the LTSH lower bank - misalignment concentrates gas flow into lanes and accelerates erosion.
- [ANALYSIS] Downstream ESP and ID fan - additional mass flow and moisture from the leak.
- [ANALYSIS] Turbine - protected in this instance because the unit was stopped in a controlled manner before the leak escalated.

5. Initial hypotheses for the failure mechanism

- H1 - Fly-ash (flue gas) erosion of the connection tubes, progressively thinning the wall until perforation. [This is the station's recorded finding.]
- H2 - Steam-jet erosion from a primary leak elsewhere in the bank, with coil 117 being a secondary victim rather than the origin.
- H3 - Long-term overheating and creep in the LTSH.
- H4 - Waterside or steam-side deposits causing local overheating.
- H5 - Corrosion - fireside or dew-point attack in the low-temperature zone.
- H6 - Weld or material defect in the connection tube joints.
- H7 - Gas-path maldistribution caused by refractory loss or a bypass lane, locally concentrating ash-laden gas onto specific coils.

5.1 Failure modes observed

- [RECORD] Failure mode: pin hole - progressive wall thinning to perforation, as distinct from a burst.
- [ANALYSIS] Mechanism class: erosive wear (wear-out failure with an increasing hazard rate), not a random or overload failure.
- [ANALYSIS] Consequential failure mode: steam-jet cutting of neighbouring coils, converting a single-tube failure into a five-coil repair.

- [ANALYSIS] Contributory mode: refractory loss at header 9 creating a gas bypass lane and locally elevating gas velocity.

6. Which evidence supports which hypothesis

- [ANALYSIS] The recorded failure type - pin hole rather than burst - is the primary discriminator. Erosion perforates a tube when the thinning wall reaches the point where the hoop stress exceeds the material strength over a very small area, producing a small hole. Overheating produces bulging and a thick- or thin-lipped longitudinal opening. The recorded morphology therefore supports H1 and argues against H3.
- [ANALYSIS] The location - LTSH lower bank connection tubes at 41 m - is a classic high-velocity, ash-laden, relatively low gas temperature zone. This supports H1 and further weakens H3, since creep requires sustained high metal temperature that this zone does not see.
- [ANALYSIS] The extent of damage spreading symmetrically across coils on both sides of the failed coil (115, 116 on one side, 118, 119 on the other) is consistent with both H1 (a broad erosion field across the bank) and H2 (jet cutting radiating from one origin). These two are separated by wall-thickness measurement of the undamaged faces, not by the damage pattern alone.
- [ANALYSIS] The station's own remedial action - fitting protection shields on the LTSH lower bank connection tubes and the economiser - is an implicit confirmation that the investigating engineers found generalised erosion rather than a single defect, since shields address a field condition.
- [ANALYSIS] Refractory being applied at header 9 as part of the remedial action supports H7: refractory loss at a header creates a gas bypass path, and the resulting local velocity increase is a recognised accelerator of tube erosion.
- [ANALYSIS] The absence of any recorded chemistry excursion, deposit finding or discoloration in the report gives no positive support to H4 or H5.

6.1 Most discriminating evidence

[ANALYSIS] The most discriminating single item is the recorded failure type of pin hole in a low-temperature convective bank. Pin-hole perforation in the LTSH lower bank connection tubes is the signature of external particle erosion and is essentially incompatible with creep, short-term overheating, or a weld defect, each of which produces a characteristically different opening. The second most discriminating item is the remedial action taken: the fitting of protection shields across a whole zone, which only makes engineering sense if the investigators found a field of thinned tubes rather than one defective tube.

7. How competing hypotheses were ruled out

- [ANALYSIS] H3 (creep / long-term overheating) ruled out - the LTSH lower bank operates at comparatively low metal temperature; the recorded failure type is a pin hole with no bulging reported; and the same erosion finding recurs across four separate events in this station within seven months, which is a wear pattern and not a temperature pattern.
- [ANALYSIS] H4 (deposits and local overheating) ruled out on the available record - no deposit, scale or bore restriction was reported at any of the 32 welded joints, and the failure was in connection tubes rather than in the highest-temperature elements.
- [ANALYSIS] H5 (corrosion) ruled out on the available record - no corrosion products, gouging or dew-point wetting were reported, and the remedial action selected (shields) is an erosion remedy, not a corrosion remedy.
- [ANALYSIS] H6 (weld or material defect) ruled out - failure occurred in tube material and the repair used standard replacement tube of the same specification; a material or weld defect would not reproduce across five coils and would not recur in the same zones in three subsequent events.
- [ANALYSIS] H2 (secondary jet cutting) cannot be fully excluded from the paperwork alone. The record names coil 117 circuit 2 as the location of failure, and the investigators clearly satisfied themselves of the origin. It is retained as the mechanism by which the damage propagated to the adjacent coils, but not as the mechanism of the initiating perforation.

- [ANALYSIS] H7 (gas maldistribution) retained as a contributing cause, evidenced by the refractory repair at header 9 being included in the remedial action.

8. Final failure mechanism determined

[RECORD] The station's determined failure mechanism is failure due to flue gas erosion. [ANALYSIS] Expanded: fly ash entrained in the flue gas, at the velocity prevailing at the 41 m elevation in the LTSH lower bank, progressively abraded the outer wall of the connection tubes. Wall thickness was reduced over an extended service period until the remaining ligament could no longer contain the internal pressure, at which point a pin-hole perforation opened. The escaping steam jet then cut the adjacent tubes on coils 115, 116, 118 and 119, which is why five coils required attention for a single initiating perforation. Refractory loss at header 9 is likely to have created a local gas bypass that raised velocity onto this specific group of coils.

9. Operational and maintenance context relevant to root cause

- [RECORD] Protection shields were provided at the LTSH lower bank connection tubes, economiser lower bank and economiser middle zone door only after this failure - meaning the erosion-prone zones were running unshielded until the tube perforated.
- [RECORD] Refractory was applied at header 9 as part of the remedial action, indicating refractory degradation had been allowed to develop.
- [RECORD] The unit was operating at 85 MW against a 250 MW rating with four mills in service - deep part-load operation.
- [ANALYSIS] Deep part-load operation with a reduced number of mills changes the gas velocity profile across the convective banks and, combined with the high excess air typically carried at low load, can raise local velocities in specific lanes even while total gas flow falls.
- [ANALYSIS] Korba coal is high in ash and in abrasive quartz content, which sets a high baseline erosion rate for this station.
- [ANALYSIS] The repair scope of 32 joints indicates that a large population of adjacent tubes was already thinned at the time of failure, i.e. the condition was extensive and had been developing undetected.
- [ANALYSIS] No record of periodic ultrasonic wall-thickness survey of the erosion-prone zones appears in the documentation supplied.

10. Final root cause

[RECORD] Root cause of failure as recorded by the station: failure due to flue gas erosion. [ANALYSIS] Converting the mechanism to a root cause: fly-ash erosion in a high-ash-coal station is a predictable, measurable wear-out mechanism with a known rate. The tube perforated not because erosion occurred, but because the erosion was neither shielded nor measured. The root cause is therefore the absence of an erosion-management regime for the low-temperature convective zones - no protection shields fitted in the known erosion-prone lanes until after the failure, no wall-thickness survey and trending to predict remaining life, and no refractory integrity check to prevent the gas-bypass lanes that concentrate the erosion.

11. Corrective actions

- [RECORD] Neighbouring damaged and eroded tube padding done.
- [RECORD] Refractory applied at header 9.
- [RECORD] Protection shields provided at LTSH lower bank connection tubes, economiser lower bank and the door of the economiser middle zone.
- [RECORD] Failure tube replaced with new straight tube 47.63 x 5.6 mm SA213 T11; 32 joints welded.
- [ANALYSIS] Additional - carry out ultrasonic wall-thickness mapping of the whole LTSH lower bank and economiser, not only the failed coils, and replace every tube below the calculated minimum allowable thickness rather than only those that have already leaked.
- [ANALYSIS] Additional - establish a thickness-trending register per coil and per elevation, so that remaining life can be predicted and replacement planned into an outage rather than forced by a trip.

- [ANALYSIS] Additional - inspect refractory at all headers in the second pass at every opportunity, since refractory loss creates the gas bypass lanes that concentrate erosion.
- [ANALYSIS] Additional - review the erosion consequences of sustained deep part-load operation with a reduced mill combination, and control excess air at low load.

Case 2 - Unit 1, 25.01.2026: Economiser upper bank coil no. 157 pin-hole leak (same trip as Case 1)

1. Asset context

Asset model and make: DSPM TPS (Dr. Shyama Prasad Mukherjee Thermal Power Station), CSPGCL, Korba East - 2 x 250 MW coal-fired units. Boiler maker reference CG-489 (as cited in the station's boiler tube leakage failure analysis letters). Natural-circulation, drum-type, pulverised-coal fired boiler with mills designated A to E, radiant and platen superheater, LTSH lower and upper banks, economiser upper and middle banks, and steam-cooled wall enclosure to the second pass. Governing mode EHG.

Operating context: Korba East fires high-ash, high-quartz Indian coal from the Korba coalfield, which is the dominant driver of fly-ash erosion in the low-temperature convective zones (economiser, LTSH lower bank and their connection tubes). All four recorded trips occurred with the unit loaded in the range 85-103 MW against a 250 MW rating, i.e. deep part load, with only 3-4 mills in service. [RECORD] Same event as Case 1: load 85 MW, drum pressure 72.59 kg/cm², main steam temperature 529 degC, mills B, C, D and E, frequency 50.06 Hz, condenser vacuum 0.0731 kg/cm². The economiser is the coldest gas-side heat transfer surface in the second pass and receives the fully developed, ash-laden gas stream.

2. Incident summary - what happened

[RECORD] In the same 25.01.2026 trip of Unit 1 at 10:00 hrs, a second zone of failure was recorded alongside the LTSH: the economiser. The location of failure was economiser upper coil no. 157, tube no. 9 from top, 8 joints. The type of failure was recorded as a pin hole and the root cause of failure as failure due to flue gas erosion. The combined repair across both zones totalled 32 welded joints with replacement straight tube of 47.63 x 5.6 mm SA213 T11. Remedial action included tube padding of neighbouring damaged and eroded tubes, refractory application at header 9, and protection shields at the LTSH lower bank connection tubes, economiser lower bank and the door of the economiser middle zone.

2.1 Timeline and onset

- [RECORD] 25.01.2026, 10:00 hrs - Unit stopped due to boiler tube leakage; two failure zones subsequently identified, LTSH and economiser.
- [ANALYSIS] The finding of two independent perforations in two different banks in one event is itself significant: it is improbable that two unrelated tubes reach end of life on the same morning by chance. The more likely reading is that both banks had reached a similar advanced state of erosion, so that the pressure test and inspection following the trip revealed a second perforation that may have been leaking already, or was on the point of doing so.
- [ANALYSIS] Onset: as with Case 1, the erosion is a long-duration wear process; the perforation is the terminal event. The 8 joints required at this location indicate a spread of damage around the failed tube.
- [RECORD] 26.01.2026, 14:00 hrs - Boiler lit up; 23 KL LDO consumed.
- [RECORD] 26.01.2026, 17:42 hrs - Unit synchronised. Outage 31.7 hours; generation loss 7.925 MU.

3. Evidence, in order of observation

- [RECORD] Two zones of failure identified in a single trip - LTSH and economiser.
- [RECORD] Economiser failure located at upper coil no. 157, tube no. 9 from top, requiring 8 joints.
- [RECORD] Failure type recorded as a pin hole.
- [RECORD] Root cause recorded as failure due to flue gas erosion.
- [RECORD] Protection shields subsequently provided at the economiser lower bank and at the door of the economiser middle zone - the investigators extended shielding to economiser zones beyond the one that failed.
- [ANALYSIS] The economiser is the lowest-gas-temperature, highest-ash-loading surface in the pass, and is the single most erosion-exposed component in a high-ash-coal boiler.

- [ANALYSIS] The same station records an economiser failure again on 11.07.2026 (Case 4), confirming a recurring zone rather than an isolated defect.

4. Components affected by the failure

- [RECORD] Economiser upper bank coil no. 157 and the tubes surrounding tube no. 9 from top.
- [RECORD] Economiser lower bank and economiser middle zone door - shielded as a precaution, therefore assessed as at risk.
- [ANALYSIS] Economiser inlet and outlet headers and their stub welds.
- [ANALYSIS] Economiser tube supports and spacers - misalignment concentrates gas lanes.
- [ANALYSIS] Downstream APH and ESP - additional moisture and mass flow while the leak persisted.
- [ANALYSIS] Feed water system - the leak represents a direct loss of feed water from the circuit.

5. Initial hypotheses for the failure mechanism

- H1 - Fly-ash erosion of the economiser tube by ash-laden gas. [Station's recorded finding.]
- H2 - Cold-end / dew-point corrosion in the coldest part of the gas path.
- H3 - Steam or water jet cutting from the concurrent LTSH leak.
- H4 - Flow-accelerated corrosion on the water side of the economiser.
- H5 - Weld or material defect at the coil.
- H6 - Gas maldistribution around the economiser middle zone door creating a local high-velocity lane.
- H7 - Ash hopper or soot-blower induced local erosion.

5.1 Failure modes observed

- [RECORD] Failure mode: pin hole - progressive external wall loss to perforation.
- [ANALYSIS] Mechanism class: abrasive wear by entrained particulate; a wear-out failure.
- [ANALYSIS] Contributory mode: gas-path bypass around the economiser middle zone door, evidenced by that door being shielded in the remedial action.
- [ANALYSIS] Consequential mode: loss of feed water into the gas path, with consequent unit stoppage.

6. Which evidence supports which hypothesis

- [ANALYSIS] The pin-hole morphology in the coldest bank supports both H1 and H2, which are the two mechanisms that produce progressive external wall loss in an economiser. They must be separated by the appearance of the wall loss - erosion produces polished, directional metal loss on the gas-facing side; dew-point corrosion produces pitted, product-covered surfaces.
- [ANALYSIS] The station's selection of protection shields as the remedy is the strongest available indication that the investigators observed erosive rather than corrosive wall loss - shields deflect particles and do nothing for acid condensation.
- [ANALYSIS] The economiser in a 250 MW unit is upstream of the air pre-heater, so its metal temperature is held well above the acid dew point by the feed water flowing through it. This substantially weakens H2 on thermodynamic grounds.
- [ANALYSIS] The physical separation between the LTSH lower bank at 41 m and the economiser upper bank makes H3 implausible - a jet from one cannot reach the other.
- [ANALYSIS] The shielding of the economiser middle zone door in the remedial action is direct support for H6, indicating the investigators identified a gas bypass at that door.
- [ANALYSIS] The recurrence of economiser erosion failures at this station within seven months weakens H5, since a material or weld defect is a one-off and would not recur in a different coil.

6.1 Most discriminating evidence

[ANALYSIS] The most discriminating evidence is the remedial action itself - the fitting of protection shields at the economiser lower bank and the middle zone door. Investigators do not fit deflector shields for corrosion, for a weld defect or for flow-accelerated corrosion; they fit them when they have seen directional particle

wear across a zone. Combined with the pin-hole morphology and the economiser's metal temperature being held above the dew point by the feed water, this fixes the mechanism as particle erosion.

7. How competing hypotheses were ruled out

- [ANALYSIS] H2 (dew-point corrosion) ruled out - the economiser tube metal is cooled by feed water and runs above the flue gas acid dew point; the cold-end corrosion zone in this boiler is the air pre-heater, not the economiser. No corrosion products were recorded.
- [ANALYSIS] H3 (jet cutting from the LTSH leak) ruled out - the two failures are in physically separate banks at different elevations; a jet cannot bridge them.
- [ANALYSIS] H4 (flow-accelerated corrosion) ruled out on the record - FAC produces internal, not external, wall loss and characteristically attacks bends and downstream of flow disturbances; nothing in the record indicates internal thinning, and the remedy chosen addresses the gas side.
- [ANALYSIS] H5 (weld or material defect) ruled out - the failure is in tube material, replacement used standard specification tube, and economiser erosion recurred in a different location six months later.
- [ANALYSIS] H7 (soot-blower erosion) not excluded by the record but not supported by it either; soot-blower erosion produces a characteristic wear pattern in a defined arc around the blower lance and would normally be identified as such. It should be checked at the next inspection.
- [ANALYSIS] H6 (gas maldistribution at the middle zone door) retained as a contributory cause on the strength of the shielding action taken at that door.

8. Final failure mechanism determined

[RECORD] Failure due to flue gas erosion. [ANALYSIS] Expanded: ash particles carried in the flue gas abraded the outer surface of economiser upper bank coil 157, tube 9 from top, over an extended service period. The wall thinned progressively until perforation. Gas bypassing around the economiser middle zone door is likely to have created a local high-velocity lane that concentrated the particle flux onto this region, which is why that door was shielded in the remedial action even though the failure itself was in the upper bank.

9. Operational and maintenance context relevant to root cause

- [RECORD] The economiser lower bank and middle zone door were shielded only after this failure, meaning these erosion-prone areas ran unprotected until a tube perforated.
- [RECORD] Two separate banks failed in the same event, indicating that the erosion condition was general across the second pass rather than local to one tube.
- [RECORD] The unit was at 85 MW against 250 MW rating with four mills - deep part load.
- [ANALYSIS] Korba coal ash loading and quartz content set a high baseline erosion rate; the erosion allowance in the original design is consumed faster than for design coal.
- [ANALYSIS] Gas bypass paths around access doors and headers are a recognised and preventable cause of localised erosion, and they develop as refractory and seals degrade.
- [ANALYSIS] Nothing in the supplied documentation indicates a periodic economiser thickness survey or an erosion-rate trend was being maintained.
- [ANALYSIS] The economiser is the most frequently failing surface in high-ash Indian units, and this is well documented industry experience - the exposure was foreseeable.

10. Final root cause

[RECORD] Failure due to flue gas erosion. [ANALYSIS] Converted to a root cause: the economiser was operated to failure against a known, predictable and measurable wear mechanism. The root cause is the absence of erosion-life management for the economiser - no baseline and periodic wall-thickness measurement, no erosion-rate trending to schedule replacement into a planned outage, no systematic sealing of gas bypass paths around doors and headers, and no shielding of the known high-velocity lanes until after a forced outage had already occurred.

11. Corrective actions

- [RECORD] Neighbouring damaged and eroded tube padding done.

- [RECORD] Protection shields provided at economiser lower bank and at the door of the economiser middle zone.
- [RECORD] Refractory applied at header 9.
- [RECORD] Failure tube replaced with new straight tube 47.63 x 5.6 mm SA213 T11.
- [ANALYSIS] Additional - carry out a full ultrasonic thickness survey of the economiser upper and lower banks at every outage, recorded coil by coil, and set a minimum allowable thickness as the replacement trigger.
- [ANALYSIS] Additional - survey and seal all gas bypass paths around access doors, headers and casing penetrations in the second pass; bypass gas is faster than the bulk stream and does disproportionate damage.
- [ANALYSIS] Additional - inspect soot blower alignment and steam pressure to confirm the blowers are not contributing to local wear.
- [ANALYSIS] Additional - consider a planned progressive replacement of the economiser bank, since two independent perforations in one event indicates the bank as a whole is approaching end of life.

Case 3 - Unit 2, 19.06.2026: LTSH lower bank coil no. 131 burst open at 41 m

1. Asset context

Asset model and make: DSPM TPS (Dr. Shyama Prasad Mukherjee Thermal Power Station), CSPGCL, Korba East - 2 x 250 MW coal-fired units. Boiler maker reference CG-489 (as cited in the station's boiler tube leakage failure analysis letters). Natural-circulation, drum-type, pulverised-coal fired boiler with mills designated A to E, radiant and platen superheater, LTSH lower and upper banks, economiser upper and middle banks, and steam-cooled wall enclosure to the second pass. Governing mode EHG.

Operating context: Korba East fires high-ash, high-quartz Indian coal from the Korba coalfield, which is the dominant driver of fly-ash erosion in the low-temperature convective zones (economiser, LTSH lower bank and their connection tubes). All four recorded trips occurred with the unit loaded in the range 85-103 MW against a 250 MW rating, i.e. deep part load, with only 3-4 mills in service. [RECORD] Parameters just before tripping: load 103 MW, drum pressure 85 kg/cm², main steam temperature 539 degC, mills in operation A, B, C and E, frequency 49.98 Hz, governor mode EHG, condenser vacuum 0.1121 kg/cm².

2. Incident summary - what happened

[RECORD] Unit 2 tripped on 19.06.2026 at 16:45 hrs, stopped due to boiler tube leakage. Generator breaker auto-tripped; GRP1 reverse power protection (S/T) 186A and 186AX, GRP2 multiplier to 186C and 186CX, GRP3 Class A trip 286A with multipliers, GRP4 Class C trip 286C with multipliers. Zone of failure: LTSH. Location of failure: LTSH lower bank coil no. 131, tube no. 3 from bottom, 2 joints, at the 41 m elevation. Type of failure: burst open. Root cause of failure: failure due to flue gas erosion. Extent of damage: adjacent LTSH lower bank coil no. 131 tube no. 4 from bottom (3 joints), coil no. 130 tube no. 4 from bottom (2 joints), coil no. 132 tube no. 3 from bottom (2 joints) and coil no. 132 tube no. 4 from bottom (2 joints). Failure tubes and bend replaced with new 47.63 x 5.6 mm SA213 T11; 11 joints welded. Boiler lit up 20.06.2026 at 14:35 hrs; synchronised at 17:45 hrs. Outage 25.00 hours; generation loss 6.250 MU; LDO consumption 16 KL and HSD consumption 16 KL.

2.1 Timeline and onset

- [RECORD] 19.06.2026, 16:45 hrs - Unit stopped due to boiler tube leakage; generator breaker auto-tripped on reverse power.
- [ANALYSIS] A burst-open failure, unlike a pin hole, releases a large steam flow immediately. The onset from first detectable symptom to the point where the unit must be taken off is correspondingly short - typically minutes to a small number of hours, against the hours-to-days available with a pin hole.
- [ANALYSIS] The failure is at the same bank, same elevation (41 m) as the January Unit 1 failure in Case 1 - a repeat of the same zone on the sister unit, five months later.
- [RECORD] 20.06.2026, 14:35 hrs - Boiler lit up. 16 KL LDO and 16 KL HSD consumed.
- [RECORD] 20.06.2026, 17:45 hrs - Unit synchronised. Outage 25.00 hours; generation loss 6.250 MU.
- [ANALYSIS] The damage pattern - the failed tube 3 on coil 131 plus tube 4 on the same coil, plus tubes on the immediately adjacent coils 130 and 132 - is a tight, symmetric cluster around the origin, characteristic of steam-jet cutting radiating from a burst.

3. Evidence, in order of observation

- [RECORD] Unit stopped at 16:45 hrs due to boiler tube leakage.
- [RECORD] Type of failure recorded as burst open - a materially different morphology from the pin holes of Cases 1, 2, 4 and 5.
- [RECORD] Location: LTSH lower bank coil 131, tube 3 from bottom, at 41 m - the same bank and elevation as the January failure on Unit 1.
- [RECORD] Root cause recorded as failure due to flue gas erosion.
- [RECORD] Damage confined to a tight cluster: the failed tube, the tube above it on the same coil, and tubes on the two immediately adjacent coils.

- [RECORD] Remedial action included refractory applied at header 4B and 5, and protection shields provided at economiser bottom zone, middle zone and LTSH lower bank tubes on the replaced tubes and bends.
- [RECORD] Replacement material 47.63 x 5.6 mm SA213 T11 - the same specification as the January repair.
- [ANALYSIS] Load 103 MW against 250 MW rating with four mills - again deep part-load operation.
- [ANALYSIS] Both LDO and HSD consumed at light-up, indicating a prolonged or difficult start relative to the LDO-only starts in the other events.

4. Components affected by the failure

- [RECORD] LTSH lower bank coils 130, 131 and 132 - tubes 3 and 4 from bottom.
- [RECORD] Refractory at headers 4B and 5.
- [RECORD] Economiser bottom and middle zones - shielded as a precaution in the same remedial action.
- [ANALYSIS] LTSH tube bends - the record specifies that bends as well as tubes were replaced, and bends are preferentially eroded because the gas stream impinges on them.
- [ANALYSIS] Tube spacers and supports in the lower bank.
- [ANALYSIS] Downstream ESP and ID fan.
- [ANALYSIS] The turbine and main steam system - protected by the controlled stoppage.

5. Initial hypotheses for the failure mechanism

- H1 - Fly-ash erosion thinning the tube wall until it burst under internal pressure. [Station's recorded finding.]
- H2 - Short-term overheating from a steam-side flow blockage in that element.
- H3 - Long-term overheating and creep rupture.
- H4 - Secondary damage - coil 131 tube 3 being cut by a jet from a primary failure elsewhere.
- H5 - Weld or material defect at the bend.
- H6 - Gas maldistribution from refractory loss at headers 4B and 5 concentrating gas velocity onto this coil group.
- H7 - Overpressure or a water-hammer event.

5.1 Failure modes observed

- [RECORD] Failure mode: burst open - a rupture, not a perforation.
- [ANALYSIS] Mechanism class: erosive wall thinning culminating in a pressure-boundary rupture when the remaining ligament falls below that required for the hoop stress.
- [ANALYSIS] Consequential failure mode: steam-jet cutting of the immediately adjacent tubes on coils 130, 131 and 132.
- [ANALYSIS] Contributory mode: refractory loss at headers 4B and 5 creating a bypass lane.

6. Which evidence supports which hypothesis

- [ANALYSIS] The burst-open morphology needs care. A burst is consistent with erosion thinning (the wall finally gives way over a larger area than a pin hole), but it is also the classic signature of overheating. The two are separated by the presence or absence of bulging and by the wall thickness of the surrounding, unfailed tube - erosion leaves a broad thinned zone with no diameter increase, overheating leaves a bulged, swollen tube.
- [ANALYSIS] The location in the LTSH lower bank - a comparatively low gas-temperature convective zone - argues strongly against H2 and H3. Creep and short-term overheat are diseases of the highest-temperature elements, not of the lower bank.
- [ANALYSIS] The main steam temperature at the time of trip was 539 degC, i.e. normal and not in excursion, which weakens the overheating hypotheses further.

- [ANALYSIS] Replacement of bends as well as tubes, and the explicit fitting of shields on the replaced tubes and bends, supports H1: bends are the impingement surfaces where erosion is fastest.
- [ANALYSIS] The refractory work at headers 4B and 5 in the same remedial action is direct support for H6.
- [ANALYSIS] The tight, symmetric damage cluster centred on coil 131 tube 3 supports that tube as the origin and weakens H4.
- [ANALYSIS] Drum pressure of 85 kg/cm² at the time of trip is a normal operating value, giving no support to H7.

6.1 Most discriminating evidence

[ANALYSIS] The most discriminating evidence is the combination of location and steam temperature: a burst in the LTSH lower bank at 41 m while main steam temperature was normal at 539 degC. Overheating mechanisms require either an abnormal temperature or a starved element in a high-temperature bank; neither applies here. That leaves wall loss as the only mechanism by which a normally pressurised, normally cooled tube can rupture, and the recurrence of the same finding across four events in the same zones identifies that wall loss as erosion. The definitive confirmation, which the paperwork does not record, would be the wall thickness measured at and around the rupture.

7. How competing hypotheses were ruled out

- [ANALYSIS] H2 (short-term overheating) ruled out - main steam temperature was normal at 539 degC, the failure is in a low-temperature bank, and no steam-side blockage was reported at the 11 welded joints.
- [ANALYSIS] H3 (creep) ruled out - the LTSH lower bank does not operate in the creep range for SA213 T11; creep failures would be expected in the platen or final superheater, not here.
- [ANALYSIS] H4 (secondary jet cutting) ruled out as the initiating failure - the damage cluster is symmetric about coil 131 tube 3 and the investigators identified that tube as the location of failure with the others recorded separately as extent of damage.
- [ANALYSIS] H5 (weld or material defect) ruled out - the recurrence of the same failure type in the same bank at the same elevation across two units in five months is inconsistent with an isolated defect.
- [ANALYSIS] H7 (overpressure or water hammer) ruled out - drum pressure at 85 kg/cm² and main steam temperature at 539 degC were both normal, and no pressure excursion is recorded.
- [ANALYSIS] H6 (gas maldistribution) retained as a contributing cause - the refractory repair at headers 4B and 5 in the same remedial action indicates the investigators found a bypass condition.

8. Final failure mechanism determined

[RECORD] Failure due to flue gas erosion. [ANALYSIS] Expanded: ash-laden flue gas at the 41 m elevation abraded the wall of LTSH lower bank coil 131, tube 3 from bottom, with the bend regions eroding fastest because the gas stream impinges directly on them. Wall thickness reduced until the remaining section could no longer carry the hoop stress at 85 kg/cm², at which point the tube burst open rather than perforating. The released steam jet then cut the adjacent tube on the same coil and tubes on coils 130 and 132. Refractory loss at headers 4B and 5 is likely to have created the local high-velocity gas lane that concentrated the erosion on this coil group.

9. Operational and maintenance context relevant to root cause

- [RECORD] Protection shields were fitted to the LTSH lower bank tubes and bends only on the replaced tubes, and to the economiser bottom and middle zones, as part of this remedial action - i.e. shielding was being extended reactively, failure by failure.
- [RECORD] Refractory required application at headers 4B and 5, showing continuing refractory degradation five months after the header 9 refractory repair in the January event.
- [RECORD] This is the second LTSH lower bank failure at the 41 m elevation in five months, the first on Unit 1 and this one on Unit 2 - the same zone is failing on both units.

- [RECORD] Deep part-load operation at 103 MW against 250 MW rating with four mills.
- [ANALYSIS] Shielding only the replaced tubes leaves the unreplaced tubes in the same bank, which have accumulated the same erosion exposure, unprotected - the remedy does not match the extent of the exposure.
- [ANALYSIS] No evidence in the record of a wall-thickness survey being used to decide how many tubes to replace; the scope appears to have been set by visible damage.
- [ANALYSIS] A repeat failure in the same zone on the sister unit is the strongest possible signal that the first event's corrective action was insufficient in scope.

10. Final root cause

[RECORD] Failure due to flue gas erosion. [ANALYSIS] Converted to a root cause: this is a repeat failure of a zone that had already failed on the sister unit five months earlier and had already been identified as erosion-prone. The root cause is therefore not merely erosion but the inadequacy of the corrective action taken after the earlier event - shielding and replacement were applied only to the tubes that had already failed or were visibly damaged, rather than to the whole exposed population identified by thickness measurement, and the same-zone exposure on the sister unit was never addressed. Underlying this is the absence of a fleet-wide erosion-life management programme covering both units.

11. Corrective actions

- [RECORD] Neighbouring damaged and eroded tube padding done.
- [RECORD] Refractory applied at header 4B and 5.
- [RECORD] Protection shields provided at economiser bottom zone, middle zone and LTSH lower bank tubes on replaced tubes and bends.
- [RECORD] Failure tubes and bend replaced with new 47.63 x 5.6 mm SA213 T11; 11 joints welded.
- [ANALYSIS] Additional - extend shielding to the entire LTSH lower bank at the 41 m elevation on both units, not only to the replaced tubes; the unreplaced tubes have the same service exposure.
- [ANALYSIS] Additional - carry out ultrasonic thickness mapping of the LTSH lower bank on Unit 1 as well, since the same zone on the sister unit has now failed and the exposure is shared.
- [ANALYSIS] Additional - measure and record the wall thickness at the rupture and around it, to confirm erosion quantitatively rather than by inference, and to establish an erosion rate in mm per 1000 hours for life prediction.
- [ANALYSIS] Additional - carry out a full refractory survey of all second-pass headers on both units rather than repairing headers one at a time as failures reveal them.

Case 4 - Unit 1, 11.07.2026: Economiser middle zone pin-hole leak at 37 m

1. Asset context

Asset model and make: DSPM TPS (Dr. Shyama Prasad Mukherjee Thermal Power Station), CSPGCL, Korba East - 2 x 250 MW coal-fired units. Boiler maker reference CG-489 (as cited in the station's boiler tube leakage failure analysis letters). Natural-circulation, drum-type, pulverised-coal fired boiler with mills designated A to E, radiant and platen superheater, LTSH lower and upper banks, economiser upper and middle banks, and steam-cooled wall enclosure to the second pass. Governing mode EHG.

Operating context: Korba East fires high-ash, high-quartz Indian coal from the Korba coalfield, which is the dominant driver of fly-ash erosion in the low-temperature convective zones (economiser, LTSH lower bank and their connection tubes). All four recorded trips occurred with the unit loaded in the range 85-103 MW against a 250 MW rating, i.e. deep part load, with only 3-4 mills in service. [RECORD] Reported by the Executive Engineer (Boiler) under letter no. 025-07/SE/BLR/BMD/U-I/4 - 2884 dated 12.07.2026, referring to boiler tube leakage on 11.07.2026 at 21:17 hrs, boiler maker CG-489, Unit 1.

2. Incident summary - what happened

[RECORD] Boiler tube leakage occurred on Unit 1 on 11.07.2026 at 21:17 hrs. Zone of failure: economiser middle at 37 m. Root cause of failure: failure due to flue gas erosion. Type of failure: pin hole. Location: steam-cooled wall LHS door bend, 2 joints. Extent of damage: adjacent door bend and straight tube, 2 joints in each. Mode of repair: failure bend and tube replaced with new 44.5 x 4.5 mm SA210 Gr A1. Number of joints: 06 Nos. Remedial action: neighbouring damaged and eroded tube padding done. Working agency M/s Cenu Engineering Works; welder Mr Trilok Kewat. Boiler light-up time 13:16 hrs on 12.07.2026; unit synchronised 17:37 hrs on 12.07.2026. Hydraulic test: not taken due to temperature constraints. Enclosure: drawing showing location of leakage.

2.1 Timeline and onset

- [RECORD] 11.07.2026, 21:17 hrs - Boiler tube leakage occurred.
- [ANALYSIS] A pin hole in the economiser zone typically announces itself through rising make-up water and falling boiler water conductivity well before the leak becomes audible; the recorded time is when the leakage was established, not necessarily when wall loss reached criticality.
- [RECORD] 12.07.2026, 13:16 hrs - Boiler lit up after repair, approximately 16 hours after the leak.
- [RECORD] 12.07.2026, 17:37 hrs - Unit synchronised.
- [RECORD] Hydraulic test not taken due to temperature constraints - the boiler was returned to service without a hydraulic pressure test of the repair.
- [ANALYSIS] This is the most consequential single entry in the record. Two days later, on 14.07.2026, the same unit tripped again on a boiler tube leakage in the platen superheater (Case 6). The decision to forgo the hydraulic test after a fast turnaround is directly relevant to the reliability of that return to service.

3. Evidence, in order of observation

- [RECORD] Boiler tube leakage on 11.07.2026 at 21:17 hrs.
- [RECORD] Zone of failure recorded as economiser middle at 37 m, while the location of failure is recorded as the steam-cooled wall LHS door bend - the failure is at the steam-cooled wall enclosure passing through the economiser middle zone elevation.
- [RECORD] Type of failure: pin hole.
- [RECORD] Root cause: failure due to flue gas erosion.
- [RECORD] Extent of damage: adjacent door bend and straight tube, 2 joints in each - damage spread to the neighbouring bend.
- [RECORD] Material replaced: 44.5 x 4.5 mm SA210 Gr A1 - carbon steel, consistent with a steam-cooled wall enclosure tube rather than an alloy superheater tube.
- [RECORD] Hydraulic test not taken due to temperature constraints.

- [ANALYSIS] The failure is at a door bend - a geometric discontinuity in the gas path where the tube deviates around an access door. These are among the most erosion-prone locations in any boiler because the gas accelerates around the obstruction.

4. Components affected by the failure

- [RECORD] Steam-cooled wall LHS door bend at the 37 m elevation in the economiser middle zone.
- [RECORD] Adjacent door bend and adjacent straight tube.
- [ANALYSIS] Access door frame, seal and surrounding refractory - the source of the gas bypass that accelerates local velocity.
- [ANALYSIS] Steam-cooled wall enclosure panel and its headers.
- [ANALYSIS] Economiser tubes in the same zone, sharing the same gas stream.
- [ANALYSIS] Downstream ESP and ID fan.
- [ANALYSIS] The repair weld joints themselves - untested by hydraulic test before return to service.

5. Initial hypotheses for the failure mechanism

- H1 - Fly-ash erosion at the door bend, accelerated by gas bypassing around the access door. [Station's recorded finding.]
- H2 - General fly-ash erosion of the steam-cooled wall unrelated to the door.
- H3 - Corrosion at the bend.
- H4 - Fatigue cracking at the bend from thermal or vibration loading.
- H5 - Manufacturing defect or thinning at the bend from the original bending operation.
- H6 - Soot-blower induced erosion.
- H7 - Overheating of the steam-cooled wall tube.

5.1 Failure modes observed

- [RECORD] Failure mode: pin hole at a bend.
- [ANALYSIS] Mechanism class: localised abrasive wear at a geometric discontinuity - the highest-erosion-rate location class in a convective pass.
- [ANALYSIS] Contributory mode: gas bypass around the access door raising local velocity; erosion rate scales steeply with gas velocity.
- [ANALYSIS] Contributory mode: bend extrados thinning from the original forming operation reduces the starting wall thickness at exactly the location of highest erosion.
- [ANALYSIS] Latent risk mode introduced by the repair: return to service without hydraulic test.

6. Which evidence supports which hypothesis

- [ANALYSIS] The failure being at a door bend rather than in a straight run is the strongest support for H1. Bends at access doors combine three erosion accelerators - direct particle impingement on the extrados, gas acceleration around the door obstruction, and reduced starting wall thickness from bend forming.
- [ANALYSIS] The extent of damage spreading to the adjacent door bend, and not to the surrounding straight tubes generally, supports the localised H1 over the general H2 - a general erosion field would thin the straight tubes as much as the bends.
- [ANALYSIS] The steam-cooled wall is by definition cooled by steam flow and operates well below creep temperature; combined with normal operating parameters, this gives no support to H7.
- [ANALYSIS] The material specified, SA210 Gr A1 carbon steel, is appropriate to the location and its use in the replacement gives no indication of a material problem, weakening H5 as a cause though bend thinning remains relevant as a contributory factor.
- [ANALYSIS] No corrosion products, cracking or fatigue indications are recorded, giving no support to H3 or H4.

- [ANALYSIS] The consistency of the finding with the other three events at this station - all attributed to flue gas erosion in the second pass - supports H1 as part of a station-wide pattern rather than an isolated cause.

6.1 Most discriminating evidence

[ANALYSIS] The most discriminating evidence is the precise location: a door bend. Every competing hypothesis would predict a different location. Corrosion would concentrate at the coldest and wettest surfaces, fatigue at restraint points and welds, general erosion across straight runs, soot-blower wear in an arc around a lance, and overheating in the hottest elements. A perforation at the extrados of a bend adjacent to an access door is the specific signature of accelerated particle erosion at a gas-path discontinuity.

7. How competing hypotheses were ruled out

- [ANALYSIS] H2 (general erosion) ruled out as the dominant mechanism - damage was confined to the door bends and their immediate straight tube, not distributed across the panel.
- [ANALYSIS] H3 (corrosion) ruled out - no corrosion products recorded, and the remedy applied (padding of neighbouring eroded tubes) is an erosion remedy.
- [ANALYSIS] H4 (fatigue) ruled out - no cracking is recorded; the failure type is a pin hole, and fatigue at a bend would present as a crack, typically circumferential or at a restraint, not as a perforation.
- [ANALYSIS] H7 (overheating) ruled out - the steam-cooled wall is a cooled enclosure surface and no temperature excursion is recorded.
- [ANALYSIS] H6 (soot-blower erosion) not positively excluded by the record; it should be checked, since the wear signature can resemble local gas-lane erosion. It is not supported by anything in the documentation.
- [ANALYSIS] H5 (bend forming thinning) retained as a contributory factor rather than a cause - it determines where erosion breaks through first but does not by itself cause the wall loss.

8. Final failure mechanism determined

[RECORD] Failure due to flue gas erosion. [ANALYSIS] Expanded: ash-laden flue gas at the 37 m elevation, accelerated as it passed the LHS access door obstruction in the economiser middle zone, impinged directly on the extrados of the steam-cooled wall door bend. The bend already had reduced wall thickness from the original forming operation. Particle abrasion removed the remaining wall progressively until a pin-hole perforation opened, and the escaping steam then damaged the adjacent door bend and straight tube.

9. Operational and maintenance context relevant to root cause

- [RECORD] Hydraulic test not taken due to temperature constraints - the repair was returned to service without pressure verification.
- [RECORD] The unit was returned to service in under 20 hours from leak to synchronisation, a fast turnaround.
- [RECORD] Remedial action was limited to padding of neighbouring damaged and eroded tubes - no shields and no refractory work are recorded for this event, unlike the January and June events.
- [ANALYSIS] The same unit tripped again on a tube leak three days later. Whether or not the two are physically connected, the sequence indicates the unit was being returned to service with an unresolved and progressing erosion condition.
- [ANALYSIS] Gas bypass around access doors is a recurring theme across these events - the January event shielded an economiser door, and this event failed at a door bend. The door regions are a systematic weak point at this station.
- [ANALYSIS] Temperature constraints preventing hydraulic testing are a real operational reality on a fast turnaround, but they mean the repair carries an unverified defect risk into service.
- [ANALYSIS] No wall-thickness measurement of the surrounding door bends is recorded, so the decision to replace only the failed and immediately adjacent bends was made on visible damage rather than on measurement.

10. Final root cause

[RECORD] Failure due to flue gas erosion. [ANALYSIS] Converted to a root cause: the immediate cause is accelerated particle erosion at a gas-path discontinuity. The root cause is that the door and bend regions - a known, repeating, station-wide weak point already identified in the January event - were not systematically surveyed, sealed or shielded across the boiler. A second and independent root cause concerns the repair process: returning the unit to service without a hydraulic test, and without a thickness survey of the surrounding bends, means the adequacy of both the repair and the remaining tube population was never verified.

11. Corrective actions

- [RECORD] Neighbouring damaged and eroded tube padding done.
- [RECORD] Failure bend and tube replaced with new 44.5 x 4.5 mm SA210 Gr A1; 06 joints.
- [ANALYSIS] Additional - survey every access door and casing penetration in the second pass on both units for gas bypass, and seal them; the bypass is what accelerates the gas onto the bend.
- [ANALYSIS] Additional - fit erosion shields to all door bends, not only to the one that failed, since they share the same exposure.
- [ANALYSIS] Additional - measure wall thickness at all accessible bends in the zone before deciding repair scope, rather than replacing only visibly damaged tubes.
- [ANALYSIS] Additional - where a hydraulic test genuinely cannot be taken because of temperature constraints, apply a compensating control: 100% radiography or dye-penetrant examination of every repair weld, recorded, and a documented engineering sign-off accepting the residual risk.
- [ANALYSIS] Additional - plan repairs so that the boiler cooling time needed for a hydraulic test is built into the outage, rather than treating the test as the item that gives way under schedule pressure.

Case 5 - Unit 1, 11.07.2026: Steam-cooled wall adjacent door bend and straight tube (extent of damage from Case 4)

1. Asset context

Asset model and make: DSPM TPS (Dr. Shyama Prasad Mukherjee Thermal Power Station), CSPGCL, Korba East - 2 x 250 MW coal-fired units. Boiler maker reference CG-489 (as cited in the station's boiler tube leakage failure analysis letters). Natural-circulation, drum-type, pulverised-coal fired boiler with mills designated A to E, radiant and platen superheater, LTSH lower and upper banks, economiser upper and middle banks, and steam-cooled wall enclosure to the second pass. Governing mode EHG.

Operating context: Korba East fires high-ash, high-quartz Indian coal from the Korba coalfield, which is the dominant driver of fly-ash erosion in the low-temperature convective zones (economiser, LTSH lower bank and their connection tubes). All four recorded trips occurred with the unit loaded in the range 85-103 MW against a 250 MW rating, i.e. deep part load, with only 3-4 mills in service. [RECORD] Same event as Case 4 - Unit 1, boiler tube leakage 11.07.2026 at 21:17 hrs, reported under letter dated 12.07.2026. This case treats the propagated damage separately from the initiating perforation, because the damage mechanism is different and requires a different remedy.

2. Incident summary - what happened

[RECORD] In the same 11.07.2026 event, in addition to the initiating pin hole at the steam-cooled wall LHS door bend, the extent of damage was recorded as the adjacent door bend and straight tube, 2 joints in each. These were replaced along with the failed bend, using 44.5 x 4.5 mm SA210 Gr A1, contributing to the total of 06 welded joints. Remedial action recorded was padding of neighbouring damaged and eroded tubes.

[ANALYSIS] The damage to these adjacent components was not itself the initiating failure; it was caused by the steam released from the initiating perforation, and by the same underlying erosion field that had thinned them in parallel with the failed bend.

2.1 Timeline and onset

- [ANALYSIS] Phase 1 - Over an extended service period, the erosion field around the LHS access door thins the failed bend and, at a similar rate, the adjacent door bend and straight tube. All are being consumed simultaneously.
- [RECORD] 11.07.2026, 21:17 hrs - The first of the group perforates; leakage occurs.
- [ANALYSIS] Phase 2 - From the moment of perforation, a high-velocity steam jet issues from the pin hole and impinges on whatever is immediately adjacent. Steam-jet cutting is rapid, measured in hours rather than in the thousands of hours the ash erosion took.
- [ANALYSIS] Phase 3 - By the time the unit was taken off and the boiler opened, the adjacent door bend and straight tube had been damaged sufficiently to require replacement, 2 joints in each.
- [RECORD] 12.07.2026, 13:16 hrs - Boiler lit up after replacement of the failed bend and the damaged adjacent components.
- [RECORD] 12.07.2026, 17:37 hrs - Unit synchronised.

3. Evidence, in order of observation

- [RECORD] Extent of damage recorded separately from the location of failure - the station's own report distinguishes the initiating tube from the damaged neighbours.
- [RECORD] The damaged components are the adjacent door bend and the adjacent straight tube - i.e. immediately neighbouring the perforation.
- [RECORD] 2 joints required in each, indicating a defined length of each had to be cut out and replaced.
- [RECORD] Remedial action of padding neighbouring damaged and eroded tubes - the word eroded in the station's own remedial action indicates the neighbours were found already worn, not merely jet-cut.

- [ANALYSIS] The fact that the adjacent door bend required replacement while more distant tubes did not is consistent with damage that falls off sharply with distance from the origin - the signature of jet impingement.
- [ANALYSIS] The adjacent components share the identical gas-path position and therefore the identical ash erosion exposure as the failed bend.

4. Components affected by the failure

- [RECORD] Adjacent steam-cooled wall door bend.
- [RECORD] Adjacent straight tube.
- [ANALYSIS] Any tube in the direct line of the escaping jet, whether or not visibly damaged - jet-cut wall loss can be present without being obvious.
- [ANALYSIS] Access door frame and seal, exposed to the escaping steam.
- [ANALYSIS] Refractory and insulation local to the leak, which is washed and degraded by the steam.

5. Initial hypotheses for the failure mechanism

- H1 - Steam-jet cutting from the initiating perforation.
- H2 - Parallel ash erosion - the neighbours were being thinned by the same gas stream and were close to failure themselves.
- H3 - The adjacent bend was in fact the initiating failure and the recorded location is the secondary one.
- H4 - Damage caused during the repair itself - cutting, grinding or handling.
- H5 - Pre-existing defects unrelated to this event.

5.1 Failure modes observed

- [ANALYSIS] Failure mode: erosive wall loss by high-velocity wet steam impingement (jet cutting) - a rapid, secondary mechanism.
- [ANALYSIS] Failure mode: parallel ash erosion of components sharing the same gas-path exposure - a slow, primary mechanism.
- [ANALYSIS] These two mechanisms act on the same components, which is why the damage extent is greater than jet cutting alone would produce in the available time.

6. Which evidence supports which hypothesis

- [ANALYSIS] The sharp fall-off of damage with distance - the immediately adjacent bend and tube affected, nothing beyond - supports H1, since a jet has a defined and limited reach.
- [ANALYSIS] The station's own use of the word eroded in describing the neighbouring tubes that were padded supports H2 as a co-acting mechanism; a purely jet-cut tube would be described as damaged, not eroded.
- [ANALYSIS] The recorded distinction between location of failure and extent of damage indicates the investigators identified an origin, which weakens H3.
- [ANALYSIS] Nothing in the record suggests repair-induced damage, giving no support to H4.
- [ANALYSIS] The combination - components already thinned by the shared erosion field, then finished off by the jet - explains why a 2-joint initiating failure produced a 6-joint total repair scope.

6.1 Most discriminating evidence

[ANALYSIS] The most discriminating evidence is the wording of the station's remedial action: padding of neighbouring damaged and eroded tubes. The distinction between damaged and eroded is meaningful. It indicates that the investigating engineers found neighbours in two conditions - some cut by the jet, and some already worn thin by the gas stream independently of this leak. That single phrase establishes that the surrounding tube population was already compromised before the perforation occurred, which is a far more serious finding than the perforation itself.

7. How competing hypotheses were ruled out

- [ANALYSIS] H3 (adjacent bend as the true origin) ruled out - the report separately identifies location of failure and extent of damage, and the investigators had the physical evidence in front of them.
- [ANALYSIS] H4 (repair-induced damage) ruled out - the damage extent was recorded as a finding, and repair damage would not be reported as extent of damage in a failure analysis letter.
- [ANALYSIS] H5 (unrelated pre-existing defects) ruled out as a separate category - the pre-existing condition was erosion from the same gas stream, which is H2 and is retained.
- [ANALYSIS] H1 and H2 both retained - they are not competing but sequential and complementary. The ash erosion prepared the components; the steam jet completed the damage.

8. Final failure mechanism determined

[ANALYSIS] Two mechanisms acting in sequence on the same components. First, ash erosion from the accelerated gas stream around the access door thinned the adjacent door bend and straight tube in parallel with the bend that eventually failed, since all three occupy the same erosion field. Second, once the initiating bend perforated, the escaping high-velocity steam jet impinged on those already-thinned neighbours and cut them further, until they too required cutting out and replacement. The recorded requirement for 2 joints in each is the measure of how much had to be removed.

9. Operational and maintenance context relevant to root cause

- [RECORD] The station's remedial action explicitly identifies neighbouring tubes as both damaged and eroded, confirming that the surrounding population was already worn.
- [ANALYSIS] Padding, which is the recorded remedy, is a weld overlay that restores thickness locally. It is a legitimate short-term measure but it does not restore the tube to original condition and it does not address the gas velocity that caused the wear.
- [ANALYSIS] Only the immediately adjacent components were replaced. Tubes slightly further from the origin, which are in the same erosion field but were not jet-cut, remain in service with unknown wall thickness.
- [ANALYSIS] No hydraulic test was taken on this repair (see Case 4), so neither the replacement joints nor the padded neighbours were pressure-verified.
- [ANALYSIS] The propagation of damage from one perforation to a 6-joint repair scope is a direct cost of late detection - the earlier a leak is detected and the unit taken off, the smaller the jet-cut damage field.

10. Final root cause

[ANALYSIS] The root cause of the propagated damage has two parts. First, the adjacent components had already lost wall thickness to the same unmanaged erosion field, so they had little margin left to survive the jet - which returns to the same root cause as Case 4, the absence of erosion survey and shielding at door and bend regions. Second, the extent of jet-cut damage is a function of how long the leak ran before the unit was taken off, which points to leak detection capability. Where an acoustic boiler tube leak detection system is either absent or not being acted upon promptly, a small perforation is allowed to become a multi-tube repair.

11. Corrective actions

- [RECORD] Neighbouring damaged and eroded tube padding done.
- [RECORD] Adjacent door bend and straight tube replaced, 2 joints in each, with 44.5 x 4.5 mm SA210 Gr A1.
- [ANALYSIS] Additional - measure wall thickness on all tubes within the jet reach and within the shared erosion field, and replace rather than pad any tube below minimum allowable thickness; padding a tube that is already below minimum thickness is a temporary measure that will fail again.
- [ANALYSIS] Additional - record which tubes were padded and where, and inspect those specific locations at the next outage, since padded tubes are known weak points.

- [ANALYSIS] Additional - review the boiler tube leak detection arrangement and the operator response protocol, since the extent of secondary damage is directly proportional to how long the leak is allowed to run.
- [ANALYSIS] Additional - treat extent of damage as an indicator in its own right: when the damage field is large relative to the initiating failure, it signals either late detection or a widely thinned tube population, and both warrant investigation beyond the failed tube.

Case 6 - Unit 1, 14.07.2026: Platen superheater inlet inner loop burst open

1. Asset context

Asset model and make: DSPM TPS (Dr. Shyama Prasad Mukherjee Thermal Power Station), CSPGCL, Korba East - 2 x 250 MW coal-fired units. Boiler maker reference CG-489 (as cited in the station's boiler tube leakage failure analysis letters). Natural-circulation, drum-type, pulverised-coal fired boiler with mills designated A to E, radiant and platen superheater, LTSH lower and upper banks, economiser upper and middle banks, and steam-cooled wall enclosure to the second pass. Governing mode EHG.

Operating context: Korba East fires high-ash, high-quartz Indian coal from the Korba coalfield, which is the dominant driver of fly-ash erosion in the low-temperature convective zones (economiser, LTSH lower bank and their connection tubes). All four recorded trips occurred with the unit loaded in the range 85-103 MW against a 250 MW rating, i.e. deep part load, with only 3-4 mills in service. [RECORD] Parameters just before tripping: load 102 MW, drum pressure 85.21 kg/cm², main steam temperature 546 degC, mills in operation A, B and C, frequency 49.93 Hz, governor mode EHG, condenser vacuum 0.0956 kg/cm². [ANALYSIS] The platen superheater is the highest-metal-temperature pressure part in the boiler and operates in the creep range for its material. This distinguishes it fundamentally from the economiser, LTSH and steam-cooled wall locations of Cases 1 to 5.

2. Incident summary - what happened

[RECORD] Unit 1 tripped on 14.07.2026 at 11:15 hrs, stopped due to boiler tube leakage. Generator breaker auto-tripped; GRP1 reverse power protection 37G1 with 186A and 186AX, GRP2 multiplier to 186C and 186CX, GRP3 Class A trip 286A with multipliers, GRP4 Class C trip 286C with multipliers. Zone of failure: platen superheater. Type of failure: burst open (entered by hand on the report). Location: platen superheater (inlet) inner loop, 2 joints. Root cause of failure: failure due to flue gas erosion. Mode of repair: failure bend replaced by new one, 51 x 7.1 mm SA213 T22; 2 joints welded. Remedial action: neighbouring damaged and eroded tube padding done. Working agency M/s Cenu Engineering Works; welder Mr Niranjan Bacchad. Boiler light-up 15.07.2026 at 14:53 hrs; unit synchronised 15.07.2026 at 20:08 hrs. Outage 32.83 hours; generation loss 8.221 MU; LDO consumption 43 KL. Hydraulic test: not taken due to temperature constraints.

2.1 Timeline and onset

- [RECORD] 11.07.2026, 21:17 hrs - Previous tube leak on the same unit, in the economiser middle zone (Cases 4 and 5).
- [RECORD] 12.07.2026, 17:37 hrs - Unit synchronised after that repair, without a hydraulic test.
- [RECORD] 14.07.2026, 11:15 hrs - Unit tripped again, approximately 42 hours after synchronising, on a fresh boiler tube leakage in the platen superheater.
- [ANALYSIS] A burst open in the platen superheater releases a large steam flow immediately; unlike a pin hole, there is little warning period. The onset is effectively instantaneous at the point of rupture, though the underlying damage accumulates over thousands of hours.
- [RECORD] 15.07.2026, 14:53 hrs - Boiler lit up; 43 KL LDO consumed, substantially more than the 16-23 KL of the other events, indicating a longer or more difficult start.
- [RECORD] 15.07.2026, 20:08 hrs - Unit synchronised. Outage 32.83 hours; generation loss 8.221 MU.
- [ANALYSIS] Two forced outages on the same unit within three days, totalling roughly 53 hours of outage and around 13 MU of generation loss, is the pattern that matters more than either event individually.

3. Evidence, in order of observation

- [RECORD] Unit stopped at 11:15 hrs due to boiler tube leakage.
- [RECORD] Zone of failure: platen superheater - the highest-temperature bank in the boiler.
- [RECORD] Location: platen superheater inlet, inner loop.
- [RECORD] Type of failure: burst open.

- [RECORD] Main steam temperature just before tripping: 546 degC - the highest of the four events recorded, and at 102 MW on a 250 MW unit.
- [RECORD] Replacement material: SA213 T22 (2.25Cr-1Mo), 51 x 7.1 mm - a creep-resistant alloy, appropriate to a high-temperature bank.
- [RECORD] Root cause recorded as failure due to flue gas erosion - the same attribution as the economiser, LTSH and steam-cooled wall failures.
- [RECORD] Only 2 joints required - a very contained repair, in contrast to the 32, 11 and 6 joints of the other events.
- [RECORD] Hydraulic test not taken due to temperature constraints.
- [RECORD] LDO consumption 43 KL, roughly double the other events.
- [ANALYSIS] The unit had been returned to service 42 hours earlier from an untested repair, though in a different zone of the boiler.

4. Components affected by the failure

- [RECORD] Platen superheater inlet inner loop bend.
- [RECORD] Neighbouring tubes, padded as remedial action.
- [ANALYSIS] Platen superheater inlet header and stub welds.
- [ANALYSIS] Platen spacers, ties and supports - loss of alignment allows platens to bow into the gas stream.
- [ANALYSIS] Adjacent platen elements exposed to the escaping steam jet.
- [ANALYSIS] Attemperator and steam temperature control system - relevant because platen inlet conditions determine the metal temperature of this element.
- [ANALYSIS] Turbine - protected in this instance by the controlled stoppage.

5. Initial hypotheses for the failure mechanism

- H1 - Fly-ash erosion of the inlet loop. [Station's recorded finding.]
- H2 - Long-term overheating and creep rupture - the classic failure mode of platen superheater tubing in SA213 T22.
- H3 - Short-term overheating from steam-side flow starvation in that loop, for example from an internal deposit or a partial blockage.
- H4 - Steam-side oxide scale exfoliation blocking the loop and starving it of cooling steam.
- H5 - Fireside corrosion from coal ash constituents at high metal temperature.
- H6 - Weld or material defect in the bend.
- H7 - Damage or disturbance introduced during the repair outage two days earlier.

5.1 Failure modes observed

- [RECORD] Failure mode: burst open at the platen superheater inlet inner loop.
- [ANALYSIS] Candidate mechanism class 1: erosive wall loss to rupture - as recorded by the station.
- [ANALYSIS] Candidate mechanism class 2: creep rupture from long-term overheating - the mechanism this location and material are most susceptible to, characterised by a thick-lipped opening, wall thinning by creep deformation, heavy internal oxide scale and no external material loss.
- [ANALYSIS] Candidate mechanism class 3: short-term overheat - characterised by a thin-lipped fish-mouth opening and significant local bulging.

6. Which evidence supports which hypothesis

- [ANALYSIS] The recorded root cause of flue gas erosion is consistent with the station's findings across all four events, and there is a genuine erosion field in this boiler, so the attribution is not unreasonable on its face.
- [ANALYSIS] However, the location weighs against it. The platen superheater sits high in the furnace where gas velocity is comparatively low and ash is partly molten or sticky rather than abrasive; it is the

economiser and LTSH lower bank, at the cold end where gas velocity is highest and ash is dry and abrasive, that are the erosion-prone zones. The other five cases in this set are all in that cold-end region. The platen is the outlier.

- [ANALYSIS] The material replaced, SA213 T22, is specified for creep resistance precisely because this location runs in the creep range. That is direct evidence that the design metal temperature here is high enough for creep to be the governing damage mechanism.
- [ANALYSIS] Main steam temperature of 546 degC at only 102 MW load is worth noting. Sustained operation at high steam temperature at low load, with only three mills in service, can produce uneven gas-side temperature distribution and locally elevated metal temperature in individual platen loops.
- [ANALYSIS] The very small repair scope of 2 joints, against 32, 11 and 6 in the erosion events, is a different signature. Erosion fields damage neighbours broadly; a single-element failure confined to 2 joints is more characteristic of a localised, element-specific mechanism such as H2, H3 or H4.
- [ANALYSIS] The inner loop of a platen inlet is a location where steam-side oxide exfoliation debris typically collects, which is the classic precondition for H4 leading to H3.
- [ANALYSIS] Nothing in the record documents wall thickness at the rupture, the presence or absence of bulging, or the internal oxide scale thickness - these are exactly the observations that would discriminate between the hypotheses, and they are absent.

6.1 Most discriminating evidence

[ANALYSIS] There is no discriminating evidence in the record for this case, and that is itself the most important finding. The three candidate mechanisms are separated by three specific observations, none of which appears in the documentation: whether the tube is bulged at the rupture (overheating) or unbulged with external material loss (erosion); the wall thickness profile around the opening, measured rather than judged; and the internal oxide scale thickness, which is thick for long-term overheating, thin for erosion and short-term overheat. Until those are recorded, the attribution of this failure to flue gas erosion is an inference by analogy with the other failures in the same boiler, not a determination supported by the evidence. Given that the platen superheater is the one location in this boiler where creep is the expected mechanism, this distinction matters considerably for what corrective action is appropriate.

7. How competing hypotheses were ruled out

- [ANALYSIS] H6 (weld or material defect) can be reasonably discounted - the repair used the correct specification and no defect is recorded, though without examination of the failed bend this is not firmly established.
- [ANALYSIS] H7 (repair-induced damage) can be discounted - the 11.07 repair was in the economiser middle zone at the steam-cooled wall, physically remote from the platen superheater inlet, and there is no plausible mechanism connecting them.
- [ANALYSIS] H5 (fireside corrosion) is not supported by anything in the record and would normally be identified visually by its characteristic laminated deposits; it is provisionally set aside pending examination of the failed bend.
- [ANALYSIS] H1 (erosion) is the station's recorded finding and cannot be ruled out - an erosion field does exist in this boiler and platen erosion does occur where ash is dry and gas velocity is locally high. It is retained.
- [ANALYSIS] H2, H3 and H4 (creep, short-term overheat, oxide exfoliation) cannot be ruled out on the available record, and the location and material make them at least as probable as H1. They are retained as live alternatives.
- [ANALYSIS] The honest position is that the competing hypotheses for this case have not been ruled out. Where the other five cases have a coherent evidence chain, this one rests on attribution by analogy.

8. Final failure mechanism determined

[RECORD] The station's determined failure mechanism is failure due to flue gas erosion. [ANALYSIS] This determination should be treated as provisional. The platen superheater inlet inner loop, in SA213 T22, is the

location in this boiler where long-term overheating and creep rupture are the expected end-of-life mechanism, and where steam-side oxide exfoliation and consequent loop starvation are a recognised cause of burst failures. The evidence needed to distinguish these from erosion - bulging, wall thickness profile, internal oxide scale thickness - is not recorded. The recommended action is metallurgical examination of the retained failed bend, if it still exists, to establish the mechanism definitively. If the mechanism is creep rather than erosion, the correct remedy is a remaining-life assessment of the platen bank, not the fitting of erosion shields.

9. Operational and maintenance context relevant to root cause

- [RECORD] The unit had been synchronised only 42 hours earlier after a separate tube leak repair, which was returned to service without a hydraulic test.
- [RECORD] This is the fourth recorded boiler tube leakage failure across the two units in under seven months, all attributed to the same root cause of flue gas erosion.
- [RECORD] Remedial action for this event was limited to padding of neighbouring damaged and eroded tubes - no shields, no refractory, no life assessment.
- [RECORD] Sustained operation at 85-103 MW against a 250 MW rating across all four events, with three to four mills in service.
- [ANALYSIS] Extended low-load operation with a reduced mill set produces uneven furnace heat release and gas-side temperature maldistribution, which can leave individual platen elements running hotter than the bulk steam temperature suggests. Steam temperature of 546 degC at 102 MW is consistent with the superheater being worked hard relative to the load.
- [ANALYSIS] Nothing in the supplied documentation indicates that platen superheater tube metal temperatures, oxide scale thickness or creep life have ever been assessed.
- [ANALYSIS] Repeated attribution of every failure to a single cause - flue gas erosion - across four events and four different zones is itself a warning sign in an investigation process. It suggests a default finding rather than a per-event determination.

10. Final root cause

[RECORD] Failure due to flue gas erosion. [ANALYSIS] Two root causes should be distinguished. If the mechanism is confirmed as erosion, the root cause is the same as Cases 1 to 5 - no erosion survey, no thickness trending, no systematic shielding. If the mechanism is creep or oxide-exfoliation-driven starvation, as the location and material make plausible, the root cause is different and more serious: the absence of any high-temperature pressure part life management at this station - no tube metal temperature monitoring, no oxide scale measurement, no remaining-life assessment on a bank that is operating in the creep range. Beyond the physical mechanism, there is a root cause in the investigation process itself. Determining the mechanism requires examining the failed component, and that examination does not appear to have been carried out for any of these four events.

11. Corrective actions

- [RECORD] Neighbouring damaged and eroded tube padding done.
- [RECORD] Failure bend replaced by new one, 51 x 7.1 mm SA213 T22; 2 joints welded.
- [ANALYSIS] Priority action - retain and metallurgically examine the failed bend: measure wall thickness at and around the rupture, record whether the tube is bulged, measure internal oxide scale thickness, and examine the microstructure for creep voids. This single examination determines which of the two very different corrective programmes below is the right one.
- [ANALYSIS] If erosion is confirmed - map wall thickness across the platen inlet loops, fit shields at the impingement locations, and investigate the local gas velocity and any bypass.
- [ANALYSIS] If creep or overheating is confirmed - carry out a remaining-life assessment of the platen bank, measure steam-side oxide scale across representative elements, review platen tube metal temperature instrumentation, and plan a bank replacement rather than repairing element by element.
- [ANALYSIS] Additional - review the effect of sustained low-load operation with a three-mill combination on gas-side temperature distribution and on platen metal temperatures.

- [ANALYSIS] Additional - apply the same compensating controls for the omitted hydraulic test as set out in Case 4: full radiographic or dye-penetrant examination of the repair welds with documented engineering acceptance.
- [ANALYSIS] Additional - change the failure investigation process so that the root cause is determined from examination of the failed component, and a per-event determination is recorded, rather than the same cause being entered for every event.

Cross-cutting findings across the six cases

[ANALYSIS] Read as a set rather than as four separate incidents, the records point to a small number of systemic issues. These are offered for the station's consideration.

A. The same zones are failing repeatedly on both units

The LTSH lower bank at the 41 m elevation failed on Unit 1 in January and on Unit 2 in June. The economiser failed on Unit 1 in January and again in July. This is not four unrelated failures; it is one degradation condition presenting four times. A repeat failure in a zone that has already been repaired is the strongest available evidence that the scope of the earlier corrective action was too narrow.

B. Corrective action is being scoped by visible damage, not by measurement

In each event, the tubes replaced were those that had failed or were visibly damaged, and shields were fitted on the replaced tubes. Nowhere in the records is there a wall-thickness measurement used to decide how many tubes to replace, or an erosion rate in mm per 1000 hours from which remaining life could be calculated. Without measurement, every repair leaves behind an unknown population of tubes at an unknown point in their erosion life, which is why the next failure is never far away.

C. Refractory and gas bypass are a recurring thread

Refractory was applied at header 9 in January, at headers 4B and 5 in June, and a failure occurred at an access door bend in July. Refractory loss and door bypass create local high-velocity gas lanes, and erosion rate rises steeply with gas velocity. Repairing refractory one header at a time as each failure exposes it is treating the symptom; a full refractory and casing-seal survey of the second pass on both units would treat the cause.

D. Hydraulic testing is being omitted under schedule pressure

Both July repairs record hydraulic test not taken due to temperature constraints. The constraint is real - the boiler has to cool before it can be filled and tested, and that costs outage hours. But the effect is that repairs are returned to service unverified. If the test genuinely cannot be taken, a compensating control is needed: full radiography or dye-penetrant examination of every repair weld, recorded, with documented engineering acceptance of the residual risk.

E. Every failure is attributed to the same root cause

All four events record failure due to flue gas erosion. For the economiser, LTSH lower bank and steam-cooled wall door bend, that attribution is well supported - those are the high-velocity, dry-ash, cold-end zones where erosion is the expected mechanism, and the remedies chosen (shields, padding) are erosion remedies. For the platen superheater inlet inner loop it is much less clear. That is the hottest bank in the boiler, tubed in SA213 T22 precisely because it operates in the creep range, and creep rupture and oxide-exfoliation-driven starvation are the expected mechanisms there. The repair scope of 2 joints, against 32, 11 and 6 elsewhere, is also a different signature. When the same cause is entered for every event across four different zones, it is worth asking whether it is being determined per event or applied as a default.

F. The single most valuable next action

Metallurgical examination of the failed platen superheater bend from 14.07.2026, if it has been retained. Three measurements settle the question: whether the tube is bulged at the rupture, the wall thickness profile around the opening, and the internal oxide scale thickness. Erosion, creep and short-term overheating give three different answers, and they call for three completely different corrective programmes - shields and thickness mapping for erosion, remaining-life assessment and bank replacement planning for creep. Fitting erosion shields to a bank that is failing by creep would consume the outage and not prevent the next failure.

G. The part-load operating regime deserves examination

All four failures occurred at 85-103 MW on a 250 MW unit with three or four mills. Sustained deep part-load operation changes gas velocity distribution across the convective banks, and with a reduced mill set it produces uneven furnace heat release and gas-side temperature maldistribution. Main steam temperature of 546 degC at 102 MW indicates the superheater was being worked hard relative to load. Whether the units are

being run in a regime the boiler was not designed for is a question worth putting alongside the tube metallurgy.